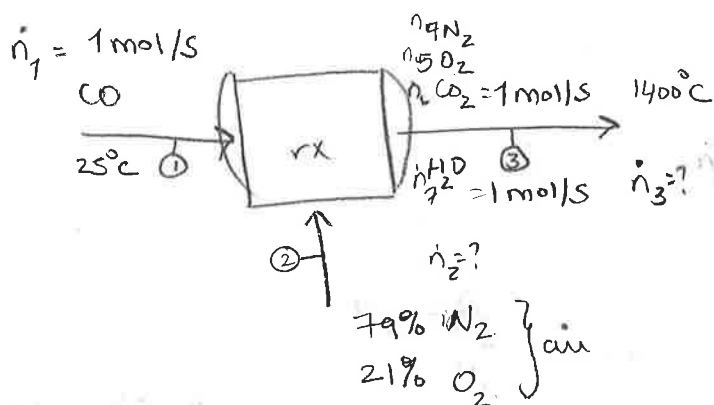
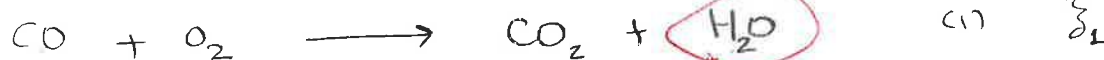


Carbon monoxide (CO) is to be burned with excess air in an adiabatic reactor. Assume 100% conversion of the CO and a basis of 1 mol/s of CO fed to the reactor. The reactants are fed at 25°C and the outlet gases exit at 1400°C. (a) Calculate the percentage excess air fed to the reactor. (b) How would increasing the percentage excess air affect the outlet temperature?



where
did this
come
from?
There's no hydrogen going
into the reactor!

$$\dot{Q} = \Delta \dot{H} = \sum \dot{Q}_L = \sum \dot{n}_e \hat{H}_e - \sum \dot{n}_i \hat{H}_i$$

$$= 1(-524.8) + 47.07($$

According to stoich:

Species Balance:

$$n_{\text{CO}} = n_{\text{O}_2} \Rightarrow n_{\text{O}_2} = 1 \text{ mol/s}$$

$$\dot{n}_{\text{O}_2} = 0.21 \dot{n}_2 \Rightarrow$$

$$\dot{n}_2 = \frac{n_{\text{O}_2}}{0.21} = \frac{1 \text{ mol/s}}{0.21} = 4.76 \text{ mol/s}$$

$$\text{N}_2: \dot{n}_{\text{N}_2, \text{out}} = \dot{n}_{\text{N}_2, \text{in}} \Rightarrow \dot{n}_{\text{N}_2, \text{out}} = 0.79 \dot{n}_2$$

$$\text{CO: } \dot{n}_{\text{CO}, \text{out}} = \dot{n}_{\text{CO}, \text{in}} - \dot{\xi}$$

$$\Rightarrow 0 = 1 \text{ mol/s} - \dot{\xi} \Rightarrow \dot{\xi} = 1 \text{ mol/s}$$

$$\text{O}_2: \dot{n}_{\text{O}_2, \text{out}} = \dot{n}_{\text{O}_2, \text{in}} - \dot{\xi}$$

$$\dot{n}_{\text{O}_2, \text{out}} = 0.21 \dot{n}_2 - 1$$

$$\text{CO}_2: \dot{n}_{\text{CO}_2, \text{out}} = \dot{n}_{\text{CO}_2, \text{in}} + \dot{\xi}$$

$$\dot{n}_{\text{CO}_2, \text{out}} = 1 \text{ mol/s}$$

$$\text{H}_2\text{O: } \dot{n}_{\text{H}_2\text{O}, \text{out}} = \dot{n}_{\text{H}_2\text{O}, \text{in}} + \dot{\xi}$$

$$\dot{n}_{\text{H}_2\text{O}, \text{out}} = 1 \text{ mol/s}$$

$$\dot{n}_{\text{N}_2, \text{out}} + \dot{n}_{\text{O}_2, \text{out}} + \dot{n}_{\text{CO}_2, \text{out}} + \dot{n}_{\text{H}_2\text{O}, \text{out}} = \dot{n}_3$$

$$\dot{n}_{\text{N}_2, \text{out}} = 0.79 \dot{n}_2$$

$$0.79 \dot{n}_2 + 0.21 \dot{n}_2 + 2 = \dot{n}_3$$

$$\dot{n}_{\text{O}_2, \text{out}} = \dot{n}_3 - 2 - 0.79 \dot{n}_2$$

$$\dot{n}_2 - \dot{n}_3 = -2$$

Table B.8 Specific Enthalpies of Selected Gases: SI Units

$\hat{H}(\text{kJ/mol})$							
Reference state: Gas, $P_{\text{ref}} = 1 \text{ atm}$, $T_{\text{ref}} =$							
T	Air	O ₂	N ₂	H ₂	CO	CO ₂	H ₂ O
0	-0.72	-0.73	-0.73	-0.72	-0.73	-0.92	-0.84
25	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100	2.19	2.24	2.19	2.16	2.19	2.90	2.54
200	5.15	5.31	5.13	5.06	5.16	7.08	6.01
300	8.17	8.47	8.12	7.96	8.17	11.58	9.57
400	11.24	11.72	11.15	10.89	11.25	16.35	13.23
500	14.37	15.03	14.24	13.83	14.38	21.34	17.01
600	17.55	18.41	17.39	16.81	17.57	26.53	20.91
700	20.80	21.86	20.59	19.81	20.82	31.88	24.92
800	24.10	25.35	23.86	22.85	24.13	37.36	29.05
900	27.46	28.89	27.19	25.93	27.49	42.94	33.32
1000	30.86	32.47	30.56	29.04	30.91	48.60	37.69
1100	34.31	36.07	33.99	32.19	34.37	54.33	42.18
1200	37.81	39.70	37.46	35.39	37.87	60.14	46.78
1300	41.34	43.38	40.97	38.62	41.40	65.98	51.47
→1400	44.89	47.07	44.51	41.90		71.89	56.25
1500	48.45	50.77	48.06	45.22	48.51	77.84	61.09

CO $\hat{n}_1 = 1$ $\hat{H}_1 = 44.95 \text{ kJ/mol}$
 O_2 $2\hat{n}_2$ $\hat{H}_2 = 47.07 \text{ kJ/mol}$
 N_2 $0.79\hat{n}_2$ $\hat{H}_3 = 44.51 \text{ kJ/mol}$
 CO_2 $\hat{n}_4$ $\hat{H}_4 = 71.89 \text{ kJ/mol}$
 H_2O $\hat{n}_5$ $\hat{H}_5 = 56.25 \text{ kJ/mol}$

$\Delta \hat{H}_r^\circ = \hat{H}_f^\circ(\text{H}_2\text{O}) + \hat{H}_f^\circ(\text{CO}_2) - \hat{H}_f^\circ(\text{CO})$
 $= -241.83 \text{ kJ/gmol} - 393.5 \text{ kJ/gmol} + 110.53 \text{ kJ/gmol}$
 $\Delta \hat{H}_r^\circ = -524.8 \text{ kJ/gmol}$